📌 BookSpot Backend Development Backlog & Progress Matrix

📊 Project Overview Status

- **Core Status:** 🟢 Auth Base Completed | 🔴 Feature Development
- **Database Engine:** PostgreSQL (Prisma ORM)
- **Cache & Message Queue:** Redis + BullMQ
- **Target Standard:** Production-Ready, High-Concurrency Safe

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🟢 Module 0: Auth & Security Base (Core Finished)

- [x] Register User (`POST /api/auth/register`)
- [x] Login User (`POST /api/auth/login`)
- [x] Access Token JWT Implementation
- [x] Refresh Token Storage
- [x] Refresh Token Rotation Mechanism
- [x] Logout (`POST /api/auth/logout`)
- [x] Authentication Middleware (`authenticate`)
- [x] Role Authorization Middleware (`authorize("ADMIN", "OWNER")`)
- [x] Global & Specific Rate Limiting (Redis Store)

🔴 Module 1: Venue Domain (`/api/venues`)

> **Tujuan:** Mengelola entity tempat (GOR, Studio, dll) dan akses kontrol Owner/Admin.

- [ ] Schema Zod Validation (`venue.schema.ts`)
- [ ] Create Venue (`POST /api/venues`) `[OWNER, ADMIN]`
- [ ] Get Venue Detail (`GET /api/venues/:id`) `[PUBLIC]`
- [ ] Update Venue (`PUT/PATCH /api/venues/:id`) `[OWNER, ADMIN]`
- [ ] Delete / Deactivate Venue (`DELETE /api/venues/:id`) `[OWNER, ADMIN]`
- [ ] List & Search Venues (`GET /api/venues`) `[PUBLIC]`
  - [ ] Filter by Type / Category
  - [ ] Filter by Location / City
  - [ ] Filter by Status (`ACTIVE` , `INACTIVE`)
  - [ ] Pagination & Geolocation Search (`lat` , `lng`)

- [ ] Venue Status Approval Workflow (`DRAFT → PENDING_APPROVAL → ACTIVE / REJECTED`) `[ADMIN]`

🔴 Module 2: Resource Domain (`/api/resources`)

> **Tujuan:** Mengelola item/unit yang benar-benar disewakan di dalam sebuah Venue.

- [ ] Schema Zod Validation (`resource.schema.ts`)
- [ ] Create Resource under Venue (`POST /api/venues/:venueId/resources`) `[OWNER, ADMIN]`
- [ ] Get Resource Detail (`GET /api/resources/:id`) `[PUBLIC]`
- [ ] Update Resource (`PATCH /api/resources/:id`) `[OWNER, ADMIN]`
- [ ] Delete / Deactivate Resource (`DELETE /api/resources/:id`) `[OWNER, ADMIN]`
- [ ] List Resources by Venue (`GET /api/venues/:venueId/resources`) `[PUBLIC]`
- [ ] Resource Pricing & Capacity Metadata Configuration

🔴 Module 3: Availability Engine (`/api/availability`)

> **Tujuan:** Mesin penghitung slot waktu dan status ketersediaan resource.

- [ ] Define Operating Hours & Slot Granularity (e.g., 60-min slot)
- [ ] Slot Generator Service (Convert operational hours to daily time slots)
- [ ] Check Availability Endpoint (`GET /api/resources/:id/availability?date=YYYY-MM-DD`) `[PUBLIC]`
- [ ] Block Resource Slot for Maintenance / Holiday `[OWNER, ADMIN]`
- [ ] Real-time Schedule Intersect Checker

🔴 Module 4: Booking Core & State Machine (`/api/bookings`)

> **Tujuan:** Alur transaksi pemesanan slot waktu oleh Customer.

- [ ] Booking State Machine Definition (`PENDING → CONFIRMED / CANCELLED / EXPIRED / COMPLETED`)
- [ ] Create Booking Endpoint (`POST /api/bookings`) `[CUSTOMER]`
- [ ] Get Booking Detail (`GET /api/bookings/:id`) `[AUTHENTICATED]`
- [ ] List Customer Bookings (`GET /api/bookings/me`) `[CUSTOMER]`
- [ ] Cancel Booking (`PATCH /api/bookings/:id/cancel`) `[CUSTOMER, OWNER]`
- [ ] List All Bookings per Venue `[OWNER, ADMIN]`
- [ ] Dynamic Booking Price Calculator

🔴 Module 5: High-Concurrency Booking Engineering ⭐ (Portfolio Hero Feature)

> **Tujuan:** Menjamin keandalan data saat terjadi lonjakan request secara bersamaan (*Race Condition Protection*).

- [ ] PostgreSQL Exclusion Constraint Setup (Preventing overlapping TSRANGE timestamps)
- [ ] Atomic Database Transaction Rollback Handling
- [ ] Redis Distributed Lock implementation (Redlock / Mutex strategy)
- [ ] Idempotency Key Handling on Booking Submissions
- [ ] Define booking expiration rules
- [ ] Concurrency Integration Test Script Execution

🔴 Module 6: Payment Engine & Mock Webhook (`/api/payments`)

> **Tujuan:** Pemrosesan transaksi keuangan dan konfirmasi pembayaran.

- [ ] Create Payment Invoice (`POST /api/payments`)
- [ ] Payment Status Verification & Expiration Tracking
- [ ] Mock Payment Gateway Integration
- [ ] Webhook Callback Receiver Endpoint (`POST /api/payments/webhook`)
- [ ] Idempotent Webhook Processing (Protecting against duplicate webhook calls)
- [ ] Automatic Booking Status Update on Success/Failure (`PENDING` → `CONFIRMED` / `EXPIRED`)

🔴 Module 7: Async Background Jobs & Workers (BullMQ + Redis)

> **Tujuan:** Mengolah proses asynchronous di belakang layar.

- [ ] BullMQ Setup & Redis Queue Connection
- [ ] Job Worker: Auto-Expire Unpaid Booking after 15 Minutes(Implement expiration using BullMQ worker)
- [ ] Job Worker: Email / Notification Dispatcher
- [ ] Job Worker: Booking Schedule Reminder Worker
- [ ] Retry Mechanism & Failed Job Queue (Dead Letter Queue)

🔴 Module 8: Notifications System (`/api/notifications`)

> **Tujuan:** Saluran komunikasi informasi transaksi ke pengguna.

- [ ] Nodemailer / Resend Provider Configuration

- [ ] In-App Notification Schema & Service
- [ ] Transactional Email Templates (HTML Engine):
  - [ ] Booking Created
  - [ ] Payment Successful
  - [ ] Booking Cancelled / Expired
  - [ ] Upcoming Booking Reminder

🔴 Module 9: Review & Rating System (`/api/reviews`)

> **Tujuan:** Sistem umpan balik pengguna setelah pemakaian fasilitas.

- [ ] Create Review Endpoint (`POST /api/bookings/:id/reviews`) `[CUSTOMER]`
- [ ] Verification Guard: Enforce `status === COMPLETED` & One Review per Booking Constraint
- [ ] Update / Delete Review `[CUSTOMER, ADMIN]`
- [ ] List Reviews by Venue / Resource `[PUBLIC]`
- [ ] Automatic Venue Average Rating Recalculator Service

🔴 Module 10: Analytical Dashboard API (`/api/dashboards`)

> **Tujuan:** Menampilkan data rekapitulasi dan metrik bisnis.

- [ ] Customer Dashboard (`/api/dashboards/customer`)
  - Upcoming Bookings, Active Invoices, History Summary
- [ ] Owner Dashboard (`/api/dashboards/owner`)
  - Revenue Metrics, Total Occupancy Rate, Popular Resources
- [ ] Admin Dashboard (`/api/dashboards/admin`)
  - Global Platform Revenue, Pending Venue Approvals, Total Active Users

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🔴 Module 11: Extended Auth & Profile Management

> **Tujuan:** Melengkapi fitur manajemen akun setelah fitur utama selesai.

- [ ] Update Profile Endpoint (`PATCH /api/users/me`)
- [ ] Change Password Endpoint (`PATCH /api/users/change-password`)
- [ ] Account Temporary Lockout Policy
- [ ] Email Verification Token Workflow
- [ ] Forgot & Reset Password Token Workflow via Email

🔴 Module 12: Automated Testing Suite

> **Tujuan:** Menjamin stabilitas kode dan performa sistem.

- [ ] **Unit Testing (Jest / Vitest):**
  - [ ] Price Calculation Logic Test
  - [ ] Slot Availability Generator Test
  - [ ] Payment Verification Service Test
- [ ] **Integration Testing (Supertest):**
  - [ ] Auth Flow (Register → Login → Auth Request)
  - [ ] Booking Flow (Search → Check Availability → Create Booking → Pay)
- [ ] **Concurrency Testing (Vitest Parallel / K6 Script) ⭐:**
  - [ ] 100 Parallel Requests hitting exact same Slot
  - [ ] Assert Outcome: **1 HTTP 201 Success** & **99 HTTP 409/400 Conflict Rejected**

🔴 Module 13: System Load & Stress Testing (k6 / Artillery)

> **Tujuan:** Mengukur batas kemampuan throughput server.

- [ ] Load Test: Search & Filter Venue Throughput
- [ ] Load Test: Availability Slot Calculation Load
- [ ] Stress Test: Payment Webhook Receiver
- [ ] Load Test Report Generation for Portfolio Case Study